



Effect of dietary inclusion of a herbal extract mixture and different oils on pig performance and meat quality



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ABSTRACT

The aim of the study was to evaluate the effect of a herbal extract mixture on pig performance and meat quality. The experiment was performed on 60 fatteners (60 ± 0.5 – 112 ± 2.0 kg). Group I (control) was fed with standard feed; groups II and III received the same feed supplemented with 150 mg BHT or 500 mg of a herbal extract mixture (sage, nettle, lemon balm and coneflower) per kg of feed, respectively. In each group, half of the animals received 4% rapeseed oil, the other half soybean oil. The herbal extracts had no effect on animal performance but significantly improved meat oxidative stability, lowered cholesterol and TI index and increased PUFA content in meat. Slight differences between animals fed with rapeseed or soybean oils were observed. Gilt meat had significantly better ($P \leq 0.01$) AI, TI, and h/H indices than barrow meat. It was concluded that herbal extracts have a beneficial effect on pork health-promoting properties due to changes in lipid fraction.

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1. Introduction

Herbs and their extracts have been used in human and veterinary medicine for a long time (Viegi, Pieroni, Guarnera, & Vangelisti, 2003). Herbs have many positive effects, of which antimicrobial and antioxidant activity is the most important (Windisch, Schedle, Plitzner, & Kroismayr, 2008). According to a European Union directive, antibiotics and chemical growth promoters have been withdrawn from farm animal feeds (Anadón, 2006); recently, herbs or herbal extracts have attracted increasing interest as an alternative feeding strategy to replace them (Hernández, Madrid, García, Orengo, & Megias, 2004). The health-related effect of plants depends on their content of active principles. Sage (*Salvia officinalis*) contains phenolics, mainly flavonoid glycosides, which exert anti-inflammatory and antioxidative activity (Lu & Foo, 2001). Lipid peroxidation is reduced by nettle (*Urtica dioica*), which also enhances the antioxidant defense system in rats (Kanter et al., 2003). Another plant with similar properties is lemon balm (*Melissa officinalis*) whose antimicrobial activity *in vitro* was found by Mahady et al. (2005). Also, purple coneflower (*Echinacea purpurea*), used previously in folk medicine, is now the subject of research and its immunologic activity established (Goel et al., 2005).

The quality of fat in animal tissue results not only from genetic background but also from feed components, especially plant-origin oils rich in polyunsaturated fatty acids (PUFA). Data available in the literature indicate that dietary fat affects the fatty acid profile of carcass lipids determining fat and meat quality (Benz et al., 2011b; Xu et al., 2010). From

the health point of view, high content of PUFA in pork is desirable; however, their influence on meat oxidative stability, shelf life and processing is undesirable. Moreover, they lead to deterioration of organoleptic quality, since compounds formed during the oxidation process have a negative effect on the meat's taste and flavor (Faustman & Cassenas, 1990; Guo et al., 2006). Fat and meat quality should thus be balanced by feed additives or supplementation.

To our knowledge, there are no reports of the combined impact of herbal extract mixture and oils rich in polyunsaturated fatty acids on growth performance and pork quality. However, these different feeding supplements could have additive or interactive effects. Therefore, a study was conducted to test the hypothesis that dietary supplementation of rapeseed or soybean oils and herbal extract mixture (sage, nettle, lemon balm and coneflower) could improve the performance and meat quality of fattening pigs. In addition, the effect on fatty acid profile, meat oxidative stability and sensory traits induced by herbal extract mixture compared to synthetic antioxidant (BHT, butylated hydroxytoluene) was also investigated.

The aim of the present experiment was to assess the effect of supplementation of feeds containing rapeseed or soybean oils with a mixture of extracts from sage, nettle, lemon balm and coneflower on the performance and meat quality of fattening pigs.

2. Material and methods

2.0.1. Animals and experimental design

The experiment was performed on 60 fattening pigs derived from Polish Landrace × Polish Large White sows mated with a Duroc × Pietrain boar. Pigs were allocated to three groups of 20 animals (10

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gilts and 10 barrows) each. Group I (control) was fed with a basic feed mixture (Table 1) with no supplement. Group II received the same mixture supplemented with 150 mg of BHT per kg. Group III received 500 mg of dried herbal extract mixture per kg of feed. The BHT (butylated hydroxytoluene, a commercially used antioxidant) was supplied by Sigma-Aldrich. The herbal extract was a mixture of water extracts from sage (*Salvia officinalis*), nettle (*Urtica dioica*), lemon balm (*Melissa officinalis*) and coneflower (*Echinacea purpurea*) in a w/w ratio of 40:20:20:20, respectively. Extracts were supplied by Phytopharm Ltd., Nowe Miasto nad Wartą, Poland. In each group half the animals (10 pigs, i.e., 5 barrows and 5 gilts) received 4% rapeseed oil, the second half soybean oil.

At the beginning of the experiment, each pig weighed about 60 ± 0.5 kg. Animals were kept in individual straw-bedded pens and fed individually twice a day with restricted feed amounts, according to body weight: from 2.8 kg/d of feed mixture at 60 kg of body weight to 3.2 kg/d at 80 kg of body weight and up. The individual body weights of all fatteners were recorded every two weeks. During the trial, animals had free access to water. At the end of the experiment all animals were slaughtered at about 112 ± 2.0 kg of body weight.

2.1. Sampling and analysis

2.1.1. Physical parameters

The quality of carcasses was evaluated according to Tyra and Žak (2012). Samples of the *longissimus thoracis* (LT) muscle were taken for analysis from the area of the last thoracic vertebrae. Meat acidity was measured 45 min and 24 h after slaughter with a pH meter equipped with a Metron OSH 12–00 electrode. Using the CIE (L*a*b*) system, the color of fresh meat and meat after 5 months of storage at –20 °C was estimated with a Minolta colorimeter. On this basis, chroma C*

and total color difference ΔE* was calculated according to MacDougall (2002):

$$C^* = (a^{*2} + b^{*2})^{1/2}$$

$$\Delta E^* = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}.$$

The water-holding capacity of meat was measured according to Grau and Hamm (1953).

2.1.2. Cholesterol and fatty acid content

Cholesterol content was estimated according to Rhee, Dutson, Smith, Hostetler, and Reiser (1982). The fatty acid profile was determined using a CP-Wax 58 capillary column (Varian BV, Middelburg, the Netherlands) (25 m, 0.53 mm, df – 1 μm, carrier gas – helium, 6 mL/min), with a column oven temperature program from 90 to 200 °C, using a Varian 3400 gas chromatograph (Varian Associates Inc., Walnut Creek, USA) equipped with a Varian 8200 CX Autosampler (200 °C), FID detector (260 °C), and Star Chromatography Workstation software (ISO, 2011) with extraction of lipids from tissues according to the method of Folch, Lees, and Stanley (1957).

2.1.3. Oxidative stability evaluation

The level of thiobarbituric acid reactive substances (TBA-RS) was analyzed after 2 weeks and 5 months of storage at –20 °C. Meat samples were prepared according to the method of Salih with modifications (Pikul, Leszczyński, & Kummerow, 1989). Lipid quality indices, i.e., atherogenic index (AI) and thrombogenicity index (TI) were calculated (Σg/100 g) according to Ulbricht and Southgate (1991):

$$AI = [(4 \times C_{14:0}) + C_{16:0}] / [n-6 \text{ PUFA} + n-3 \text{ PUFA} + \text{MUFA}]$$

$$TI = [C_{12:0} + C_{14:0} + C_{16:0} + C_{18:0}] / [(0.5 \times \text{MUFA}) + (0.5 \times n-6 \text{ PUFA}) + (3 \times n-3 \text{ PUFA}) + n-3 / n-6 \text{ PUFA}]$$

The hypocholesterolemic (h) to hypercholesterolemic (H) ratio (h/H) was calculated according to Fernández et al. (2007):

$$h/H = (C_{18:1} + C_{18:2} + C_{18:3} + C_{20:4} + C_{20:5} + C_{22:6}) / (C_{14:0} + C_{16:0}).$$

The peroxidisability index (PI) was calculated (Σg/100 g) according to Arakawa and Sagai (1986):

$$PI = (\% \text{ monoenoic} \times 0.025) + (\% \text{ dienoic} \times 1) + (\% \text{ trienoic} \times 2) + (\% \text{ tetraenoic} \times 4) + (\% \text{ pentaenoic} \times 6) + (\% \text{ hexaenoic} \times 8).$$

The iodine value (IV) of fat was calculated using the following equation (AOCS, 1998):

$$IV(g/100 \text{ g}) = (C_{16:1} \times 0.95 + (C_{18:1} \times 0.86 + (C_{18:2} \times 1.732 + (C_{18:3} \times 2.616 + (C_{20:1} \times 0.785 + (C_{22:1} \times 0.723.$$

2.1.4. Antioxidant capacity assessment

The antioxidant activity of extracts was estimated with the *in vitro* spectrophotometric method using synthetic ABTS radical cation assay (Re et al., 1999). Results were expressed as TEAC (Trolox Equivalent Antioxidant Capacity). The content of phenolic compounds was analyzed spectrophotometrically according to Singleton and Rossi (1965) using the Folin–Ciocalteu reagent. Results were expressed as a GAE (Gallic Acid Equivalent).

2.1.5. Sensory analysis

The sensory evaluation of meat after 5 months of storage at –20 °C was made on a 5-point scale (1: poorest, 5: best). Samples were thawed at +4 °C, cut into slices approximately 30 mm thick and boiled in 0.6%

Table 1
Ingredients, chemical composition and fatty acid composition of basal feeds supplemented with different oils.

	Mixtures containing rapeseed oil	Mixtures containing soybean oil
<i>Ingredients (g/kg)</i>		
Barley, ground	628.1	628.1
Wheat, ground	100	100
Soybean meal	160	160
Wheat bran	50	50
Rapeseed oil	40	–
Soybean oil	–	40
Dicalcium phosphate	2.7	2.7
Limestone	11	11
Salt	2.2	2.2
Vitamin–mineral premix ^a	5	5
L-lysine	1	1
<i>Chemical composition</i>		
Metabolizable energy (MJ/kg)	13.2	13.2
Crude protein (g/kg)	154	154
Lysine (g/kg)	8.42	8.42
Methionine + Cystine (g/kg)	5.41	5.41
<i>Fatty acids composition (% of total fatty acids)</i>		
C14	0.30	0.38
C16	13.72	16.28
C18	2.64	2.83
C18:1	43.77	31.07
C18:2	31.61	44.85
C18:3	6.83	3.99
C20	0.32	0.32
C22:1	0.38	0.16
C22:2	0.43	0.12

^a Premix in 1 kg: Vitamin A – 1,000,000 IU; Vitamin D₃ – 200,000 IU; Vitamin E – 7.0 g; Vitamin K₃ – 0.15 g; Vitamin B₁ – 0.15 g; Vitamin B₂ – 0.4 g; Vitamin B₆ – 0.3 g; Vitamin B₁₂ – 0.002 g; pantothenic acid – 1.0 g; choline chloride – 20 g; biotin – 0.01 g; folic acid – 0.2 g; nicotinic acid – 2.0 g; manganese – 4 g; iodine – 0.08 g; zinc – 8 g; iron – 10 g; copper – 4 g; cobalt – 0.04 g; selenium – 0.02 g.

Table 2
Antioxidant activity and polyphenols content in dried herbal extract.

Item	Antioxidant activity	Content of polyphenols
	TEAC ^a	GAE ^b
Herbal extracts mixture (w/w 40:20:20)	1917 ± 86.6	43.4 ± 0.88
Sage extract (<i>Salvia officinalis</i>)	2528 ± 96.8	59.0 ± 0.86
Lemon balm (<i>Melissa officinalis</i>)	1300 ± 31.9	36.5 ± 0.86
Nettle extract (<i>Urtica dioica</i>)	743 ± 18.8	23.3 ± 0.88
Coneflower (<i>Echinacea purpurea</i>)	601 ± 16.8	16.8 ± 0.86

^a TEAC — μmol of trolox/g herbal extract.^b GAE — mg gallic acid equivalent/g herbal extract.

NaCl solution to an internal temperature of +80 °C. After cooking, the meat was left for 2 min in order to even out the temperature between its layers (Baryłko-Pikielna, 1975). Then, the slices were cut into smaller pieces and presented to the panelists. The evaluation board consisted of 6 trained individuals, who evaluated the odor, taste, tenderness and juiciness of the meat.

2.2. Statistical analysis

A statistical analysis of treatment effects was performed by three-way analysis of variance, including the experimental factors of antioxidant, oil and sex. Comparison of means proceeded with Duncan's multiple range test at levels of significance of $P \leq 0.05$ and $P \leq 0.01$. All analyses were conducted using the Statistica 10 package (StatSoft, 2011). All procedures included in the experiment were in accordance with the Second Local Cracow Ethics Committee for Experiments with Animals in Cracow, Poland.

3. Results

3.1. Characteristic of supplements

Regarding fatty acid composition of oil supplements, the rapeseed oil mixture contained more oleic and linolenic acids, but less linoleic acid, than the soybean oil mixture (Table 1). Also, the rapeseed oil was characterized by doubling the content of C_{22:1} and more than tripling the content of C_{22:2} fatty acids.

Sage extract exerted the highest antioxidant activity (2528 TEAC) and was characterized by the highest total content of phenolic compounds (59.0 GAE). Both of these values were lowest (601 TEAC and 16.8 GAE) in the coneflower extract. For the prepared extract mixture, antioxidant activity measured 1917 TEAC, phenolic content 43.4 GAE (Table 2).

3.2. Pig performance indicators

Both antioxidant additives, BHT and herbs, had no significant effect on animal daily body weight gains which in all groups ranged from 868 to 918 g. Feed efficiency and meat content in carcass were also similar in all groups (Table 3). Sex had a significant ($P \leq 0.01$) effect on carcass characteristics. Gilt carcasses contained more meat and were characterized by larger loin eye areas, while barrows possessed thicker backfat.

3.3. Meat quality traits

The oxidative stability of meat after 5 months of frozen storage was significantly ($P \leq 0.05$) improved by herbal extracts (Table 4). This improvement is shown by a lower TBA-RS when compared with the control (by 12%) and BHT (by 13%) groups. Both preparations lowered cholesterol content in meat when compared to the control group, but this difference was significant ($P \leq 0.05$) only in the case of herbal extracts. Meat acidity and water-holding capacity were unchanged by dietary treatment.

Meat lightness and redness in fresh samples and after storage were similar in all groups, but yellowness was much lower ($P \leq 0.05$) when the herbal extract was used than in animals receiving BHT. Greater color stability (lower ΔE) was found in meat of pigs fed with herbal extract than in controls ($P \leq 0.05$). Barrow meat was significantly brighter ($P \leq 0.05$) after storage than that of gilts.

Meat from animals fed with BHT presented the best results of sensory evaluation of odor and taste, but these differences were significant ($P \leq 0.01$) only when compared with the control group (Table 5). These indices were also better ($P \leq 0.01$) in pigs receiving rapeseed oil, which also positively influenced meat tenderness and juiciness. There was no significant effect of sex on meat sensory traits.

Intramuscular fat of pigs fed herbal extracts contained more PUFA ($P \leq 0.01$) than that from control and BHT groups (Table 6). The fat of pigs fed with rapeseed oil contained more ($P \leq 0.01$) linolenic, eicosapentaenoic and docosahexaenoic acids than that of pigs receiving soybean oil. Sex also affected meat fatty acid pattern and gilts contained more polyunsaturated acids ($P \leq 0.01$). The intramuscular fat of all animals was characterized by a low iodine value, although the higher content of PUFA in the meat of pigs receiving the herbal extract and in gilts increased this value significantly. Neither of the dietary oils used affected the degree of unsaturation of intramuscular fat.

Pigs receiving herbal extracts had significantly higher PUFA/SFA ratios compared to control ($P \leq 0.05$) and BHT-supplemented pigs ($P \leq 0.01$). They also had significantly ($P \leq 0.05$) lower TI indices than those in both remaining groups. There were no significant differences in AI index and h/H ratio. The meat of pigs receiving rapeseed oil was characterized by significantly higher $n-6/n-3$ ratios ($P \leq 0.01$). Gilts meat had significantly ($P \leq 0.01$) lower AI and TI indices and a higher

Table 3
Body weight gains, feed utilization and indices of carcass quality evaluation.

Indices	Experimental group (G)			Kind of oil (O)		Sex (S)		SEM ^a	P-value			I ^b
	Control	BHT	Herbal extracts mixture	Rapeseed	Soybean	Gilts	Barrows		G	O	S	
Average daily weight gains (g)	868	901	918	889	902	875	916	11.45	0.23	0.57	0.09	NS
Average feed conversion per body weight gain (kg)	3.55	3.38	3.34	3.45	3.41	3.52	3.34	0.05	0.13	0.63	0.06	NS
Cold dressing yield (%)	78.90	78.50	78.50	78.60	78.70	78.70	78.60	0.15	0.27	0.56	0.81	NS
Carcass meatness (%)	51.98	52.29	53.02	52.22	52.63	53.63 ^B	51.23 ^A	0.36	0.96	0.84	0.0002	NS
Average thickness of backfat (cm)	2.26	2.31	2.18	2.30	2.21	2.13 ^A	2.37 ^B	0.04	0.48	0.29	0.006	GxO*
Backfat thickness in point C (cm)	1.03	1.04	0.94	1.02	0.99	0.88 ^A	1.12 ^B	0.03	0.21	0.51	0.0002	GxO*
Meat of primal cuts (kg)	22.27	22.38	22.18	22.12	22.43	22.96 ^B	21.59 ^A	0.20	0.90	0.38	0.0003	GxO*
Loin eye area (cm ²)	57.05	56.79	59.64	56.98	58.67	60.38 ^B	55.28 ^A	0.76	0.42	0.29	0.001	NS

^{A,B} — mean values in the same row with different letters differ significantly at $P \leq 0.01$.^a Standard error of the mean.^b Interaction; NS — not significant.* $P \leq 0.05$.

Table 4Meat quality traits and thiobarbituric acid reactive substances values of *Longissimus thoracis* estimated after slaughter and 5 months of frozen storage.

Indices	Experimental group (G)			Kind of oil (O)		Sex (S)		SEM ¹	P-value			I ²
	Control	BHT	Herbal extracts mixture	Rapeseed	Soybean	Gilts	Barrows		G	O	S	
pH 45 min after slaughter	6.29	6.39	6.38	6.39	6.32	6.39	6.32	0.03	0.51	0.43	0.41	NS
pH after 24 h cooling	5.63	5.59	5.54	5.61	5.57	5.58	5.60	0.02	0.16	0.27	0.51	GxS [*] ; GxOxS ^{**}
Water holding capacity index (%)	25.00	24.19	24.09	25.07	23.78	24.34	24.52	0.48	0.70	0.18	0.85	NS
TBA-RS after 24 h cooling (mg/kg)	0.270	0.270	0.258	0.270	0.262	0.267	0.265	0.003	0.73	0.90	0.90	OxS [*]
TBA-RS after 5 months (mg/kg)	0.548 ^b	0.551 ^b	0.477 ^a	0.529	0.521	0.530	0.521	0.01	0.05	0.77	0.74	NS
Meat color analysis												
After 24 h cooling:												
Lightness L [*]	49.67	50.64	50.21	49.93	50.41	49.66	50.68	0.36	0.56	0.52	0.18	OxS [*]
Redness a [*]	15.99	16.33	15.83	16.00	16.10	16.19	15.90	0.11	0.20	0.65	0.23	NS
Yellowness b [*]	1.86 ^a	2.55 ^b	1.75 ^a	1.92	2.17	2.10	2.00	0.14	0.05	0.40	0.73	NS
Chroma C [*]	16.11 ^{ab}	16.57 ^b	15.93 ^a	16.12	16.29	16.37	16.04	0.11	0.04	0.51	0.20	NS
After 5 months:												
Lightness L [*]	49.46	50.46	49.38	49.72	49.81	49.90 ^a	50.63 ^b	0.42	0.46	0.91	0.04	OxS [*]
Redness a [*]	16.06	15.91	15.04	15.60	15.74	15.57	15.77	0.19	0.16	0.75	0.66	NS
Yellowness b	6.06 ^b	6.24 ^b	4.88 ^a	5.87	5.59	5.65	5.81	0.24	0.05	0.56	0.74	OxS ^{**}
Chroma C [*]	17.21 ^b	17.16 ^b	15.90 ^a	16.76	16.75	16.61	16.90	0.22	0.05	0.96	0.62	OxS [*]
ΔE	4.84 ^b	4.65 ^{ab}	3.90 ^a	4.63	4.30	4.45	4.48	0.17	0.02	0.35	0.94	OxS [*]

^{a,b} — mean values in the same row with different letters differ significantly at $P \leq 0.05$.¹ Standard error of the mean.² Interaction; NS — not significant.^{*} $P \leq 0.05$ ^{**} $P \leq 0.01$.

h/H index compared to barrow meat. PI index was also significantly ($P \leq 0.01$) higher in gilt meat.

4. Discussion

Data concerning the growth-promoting efficacy of preparations of plant origin in pig feeding varies widely. This efficacy can depend on plant species and animal housing conditions. According to Mosenthin and Zimmermann (2000), feed supplements such as probiotics and prebiotics are not effective when healthy animals are kept in good conditions and are not exposed to stress and infections. Also, they are more effective in young animals with incompletely developed digestive tract (Castillo et al., 2006). Experiments with pure sage extract reported improvement in weight gains of fatteners and meat oxidative stability (Hanczakowska & Świątkiewicz, 2005; Hanczakowska, Świątkiewicz, & Szweczyk, 2007a). Numerically higher (of above 5%) weight gains, though not confirmed statistically, were observed as well in the present experiment in pigs receiving a mixture of herbal extracts. In a study by Marcin, Lauková, and Mati (2006), extracts of sage and oregano also significantly improved the body weight gains of piglets suffering from diarrhea. Thus, it can be established that the growth-promoting effect of herbs is also due to their antimicrobial activity (Hernández et al., 2004).

According to Gatlin, See, Hansen, Sutton, and Odle (2002), the fatty acid profile of fat deposited in pigs is similar to that given in the diet, although according to López-Bote and Rey (2001) this concerns primarily adipose tissue. In the present study, regardless of dietary treatment, the

iodine value of intramuscular fat (58–60) was lower than the average for pork (68) and much lower than the upper limit appropriate for meat shelf life and processing (Benz et al., 2010, 2011a).

In the current experiment, the herbal extract significantly increased the content of PUFA, creating meat with beneficial health properties but susceptible to oxidation, but at the same time the herbal additive effectively reduced this process, which was confirmed by a significantly lower TBA-RS level after storage. Lipids are particularly susceptible to oxidation; therefore it can be assumed that herbal extracts protect unsaturated fatty acids from oxidative damage. The meat of fatteners receiving extracts contained more of these acids. An increased content of unsaturated fatty acids in pig meat due to the supplementation of feed with plant extracts was also found by Habeanu et al. (2009). According to Wood and Enser (1997), antioxidant supplements in fatteners' feed can improve the meat's fatty acid pattern and prevent oxidative changes occurring during storage and cooking. However, the BHT supplement had no such effect in the current experiment. Phenolic compounds are the main active components of herb preparations (Burt, 2004) and are mainly responsible for their antioxidant activity (Wei & Shibamoto, 2007). Sage is a popular ingredient of dietary supplements due to its high antioxidant activity (Lu & Foo, 2001), which was also found in the present experiment. Also, in an experiment by Hanczakowska et al. (2007a), extended oxidation stability of meat due to a dietary sage extract supplement was observed. A similar but non-significant improvement in meat stability was obtained using nettle extract (Hanczakowska, Świątkiewicz, & Szweczyk, 2007b). Improved

Table 5

Sensory evaluation indices of meat after thermal treatment.

Indices	Experimental group (G)			Kind of oil (O)		Sex (S)		SEM ¹	P-value			I ²
	Control	BHT	Herbal extracts mixture	Rapeseed	Soybean	Gilts	Barrows		G	O	S	
Odor	4.51 ^A	4.81 ^B	4.55 ^B	4.74 ^B	4.51 ^A	4.59	4.66	0.03	0.0002	0.0008	0.24	NS
Taste	4.52 ^A	4.79 ^B	4.63 ^{AB}	4.78 ^B	4.52 ^A	4.64	4.66	0.03	0.003	0.0000	0.72	GxO [*]
Tenderness	4.52	4.58	4.58	4.58	4.53	4.60	4.52	0.04	0.74	0.49	0.25	GxS [*] ; Oxs [*]
Juiciness	4.48	4.62	4.53	4.62 ^b	4.47 ^a	4.58	4.50	0.04	0.33	0.05	0.30	NS

^{a,b} — mean values in the same row with different letters differ significantly at $P \leq 0.05$.^{A,B} — mean values in the same row with different letters differ significantly at $P \leq 0.01$.¹ Standard error of the mean.² Interaction; NS — not significant.^{*} $P \leq 0.05$.

Table 6

Cholesterol content (mg/100 g), fatty acid composition (g/100 g of detected fatty acids), partial sums of fatty acids and related indices of *Longissimus thoracis* muscle of fattening pigs.

	Experimental group (G)			Kind of oil (O)		Sex (S)		SEM ¹	P-value			I ²
	Control	BHT	Herbal extracts mixture	Rapeseed	Soybean	Gilts	Barrows		G	O	S	
Cholesterol	56.38 ^b	55.18 ^{ab}	53.73 ^a	54.78	55.42	55.42	54.78	0.55	0.05	0.52	0.51	GxS*
Fatty acid composition												
C12:0	0.118	0.114	0.109	0.113	0.116	0.108 ^a	0.120 ^b	0.03	0.28	0.55	0.02	NS
C14:0	1.68	1.64	1.63	1.69	1.62	1.64	1.65	0.04	0.85	0.40	0.90	NS
C16:0	28.31	28.42	28.00	28.33	28.15	27.59 ^A	28.90 ^B	0.16	0.53	0.55	0.0001	NS
C16:1	2.91	3.10	2.98	2.99	3.01	2.91	3.08	0.06	0.46	0.86	0.19	NS
C18:0	11.18 ^b	11.11 ^b	10.52 ^a	10.90	10.98	10.79	11.08	0.12	0.04	0.74	0.22	NS
C18:1	39.60 ^b	40.16 ^b	37.54 ^a	39.47	38.73	38.54	39.66	0.41	0.02	0.34	0.16	NS
C18:2	11.42 ^A	11.00 ^A	13.78 ^B	11.51	12.62	13.02 ^B	11.11 ^A	0.40	0.006	0.13	0.01	NS
C18:3 γ	0.096 ^{Bb}	0.065 ^{Aa}	0.121 ^{Bc}	0.104 ^b	0.084 ^a	0.103 ^b	0.084 ^a	0.01	0.0000	0.02	0.03	NS
C18:3	0.692 ^B	0.623 ^A	0.715 ^B	0.758 ^B	0.595 ^A	0.683	0.671	0.02	0.004	0.0000	0.60	GxO*;Gx
C20:0	0.166 ^B	0.175 ^B	0.139 ^A	0.155	0.164	0.160	0.159	0.01	0.002	0.26	0.87	OxS*
C20:4	2.71 ^a	2.53 ^a	3.40 ^b	2.87	2.89	3.34 ^B	2.43 ^A	0.14	0.02	0.93	0.001	OxS*
C20:5	0.143 ^{ABa}	0.118 ^{Aa}	0.185 ^{Bb}	0.180 ^B	0.118 ^A	0.162	0.136	0.09	0.002	0.0001	0.08	NS
C22:1	0.015	0.008	0.009	0.013	0.008	0.012	0.009	0.01	0.09	0.10	0.33	NS
C22:6	0.086 ^A	0.037 ^B	0.078 ^A	0.082 ^B	0.053 ^A	0.087 ^B	0.047 ^A	0.01	0.0001	0.002	0.0000	NS
Partial sums of fatty acids												
MUFA	42.53 ^{ab}	43.28 ^b	40.53 ^a	42.47	41.75	41.47	42.76	0.44	0.03	0.39	0.13	NS
PUFA	15.14 ^A	14.37 ^A	18.29 ^B	15.51	16.36	17.39 ^B	14.47 ^A	0.55	0.006	0.40	0.005	NS
PUFA n6	14.22 ^{ABa}	13.59 ^{Aa}	17.31 ^{Bb}	14.49	15.60	16.46 ^B	13.62 ^A	0.54	0.007	0.26	0.005	NS
PUFA n3	0.921 ^B	0.778 ^A	0.978 ^B	1.020 ^B	0.765 ^A	0.931 ^b	0.854 ^a	0.02	0.0001	0.0000	0.03	NS
UFA	57.67	57.65	58.81	57.98	58.11	58.86 ^B	57.23 ^A	0.25	0.08	0.78	0.001	NS
SFA	41.47	41.44	40.40	41.19	41.04	40.30 ^A	41.94 ^B	0.26	0.12	0.75	0.001	NS
Fatty acid ratios												
PUFA/SFA	0.37 ^{ABa}	0.35 ^{Aa}	0.45 ^{Bb}	0.40	0.38	0.44 ^B	0.35 ^A	0.02	0.01	0.36	0.002	NS
n – 6/n – 3	15.76 ^a	17.62 ^{ab}	18.16 ^b	14.18 ^A	20.19 ^B	17.99 ^B	16.37 ^a	0.56	0.03	0.0000	0.04	NS
h/H	1.83	1.83	1.89	1.84	1.86	1.92 ^B	1.78 ^A	0.02	0.37	0.55	0.0007	NS
Indices												
AI	0.61	0.61	0.59	0.61	0.60	0.58 ^A	0.62 ^B	0.01	0.35	0.55	0.007	NS
TI	1.40 ^b	1.41 ^b	1.32 ^a	1.38	1.38	1.32 ^A	1.42 ^B	0.02	0.03	0.99	0.0008	NS
PI	26.4 ^{ABa}	24.6 ^{Aa}	31.8 ^{Bb}	27.7	27.5	30.7 ^B	24.6 ^A	1.02	0.007	0.91	0.002	NS
Iodine value	58.42 ^A	58.18 ^A	60.86 ^B	58.71	59.59	60.26 ^B	58.04 ^A	0.40	0.007	0.24	0.004	NS

^{a,b} – mean values in the same row with different letters differ significantly at $P \leq 0.05$; ^{A,B} – mean values in the same row with different letters differ significantly at $P \leq 0.01$.

SFA – sum of saturated fatty acids; UFA – sum of unsaturated fatty acids; MUFA – sum of monounsaturated fatty acids; PUFA – sum of polyunsaturated fatty acids; h/H – hypocholesterolemic/Hypercholesterolemic ratios; AI – atherogenic index; TI – thrombogenicity index; PI – peroxidizability index.

¹ Standard error of the mean.² Interaction; NS – not significant.* $P \leq 0.05$.

oxidative stability of meat could be the reason for smaller changes in meat color. These results, however, are not consistent with those obtained by Simitzis, Symeon, Charismiadou, Bizelis, and Deligeorgis (2010), Janz, Morel, Wilkinson, and Purchas (2007) and Haak, Raes, Van Dyck, and De Smet (2008), perhaps because single preparations were used in these experiments. Probably the mixture of active substances used in the present experiment produced a more apparent synergistic effect.

Apart from improving meat oxidative stability herbal extracts also reduce cholesterol content (Fermont, Gozzelino, & Linard, 2000). This activity of herbal extracts was observed in the current experiment, as well as in earlier studies (Hanczakowska et al., 2007a,b), in meat but not in blood serum. The higher PUFA/SFA ratio in the meat of fatteners receiving herbal extracts also indicates the latter's health-promoting effects; on the other hand, they significantly increased the n – 6/n3 ratio. According to recommended optima, this latter ratio should be 1:1 to 4:1 (Simopoulos, 2004), though in European diets it ranges from 15:1 to almost 17:1 (Fernández et al., 2007). Thus, in the meat of animals from both experimental groups, the n6/n3 ratio is above this limit. A lower n – 6/n – 3 ratio (10:2) was found by Razmaite and Švirmickas (2012) in intramuscular fat of pigs, while the indices (AI, TI and h/H) were similar to those obtained in the current experiment.

Changes in the fatty acid pattern of intramuscular fat could be one of the reasons for improved odor and taste. Another possible reason is the antimicrobial activity of plant extracts, which prevents microbial decarboxylation of amino acids and production of biogenic amines (Windisch et al., 2008) and/or skatole (Lanthier, Lou, Ternier, & Squires, 2006). The presence of these compounds in meat of pigs can change its sensory quality (Claus, Lösel, Lacorn, Mentschel, & Schenkel, 2003).

5. Conclusions

To summarize the results, it can be concluded that extracts from plants evaluated in this experiment did not change pig performance, but improved the fatty acid pattern, oxidative stability and odor and taste of meat. They also reduced the cholesterol content of meat and reduced atherogenic indices.

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